

TIMING REPORT :-

Parameter	Value	Status
Tool	OpenSTA 2.5.0	-
Design Name	alu	-
Technology	Generic 45 nm Std Cell	-
Process Corner	TT (Typical)	-
Voltage	1.0 V	-
Temperature	25° C	-
Clock Name	clk	-
Clock Period	10.000 ns	✓
Clock Frequency	100 MHz	✓
Clock Uncertainty	0.100 ns	✓
Clock Latency	0.200 ns	✓
Data Arrival Time	6.845 ns	✓
Required Arrival Time	9.900 ns	✓
Setup Slack	+3.055 ns	PASS
Hold Slack	+0.420 ns	PASS
Worst Negative Slack (WNS)	0.000 ns	PASS
Total Negative Slack (TNS)	0.000 ns	PASS
Setup Violations	0	PASS
Hold Violations	0	PASS
Overall Timing Status	PASSED	✓

VIOLATION TIMING :-

Parameter	Value	Status
Clock Period	5.000 ns	-
Data Arrival Time	5.380 ns	-
Required Arrival Time	5.000 ns	-
Setup Slack	-0.380 ns	FAIL
Hold Slack	+0.120 ns	PASS
Worst Negative Slack (WNS)	-0.380 ns	FAIL
Total Negative Slack (TNS)	-1.420 ns	FAIL
Violating Paths	4	FAIL
Overall Timing Status	FAILED	✗

How to Fix Timing Violations

If setup violations occur (negative setup slack):

- **Pipeline the datapath** by adding registers to shorten combinational paths.
- **Reduce logic depth** by simplifying arithmetic or logic expressions.
- **Increase the clock period** (lower the operating frequency) if performance requirements allow.
- **Re-synthesize with higher optimization effort** (e.g., abc or timing-driven optimization).
- **Use faster standard cells** from the timing library (where available).

If hold violations occur (negative hold slack):

- Insert delay buffers on short data paths.
- Increase hold margins using synthesis or P&R optimization.
- Adjust clock skew through clock-tree synthesis.